ISLR 6.6.5

Let $x = x_{11} = x_{12} = -x_{21} = -x_{22}$, $y = y_1 = -y_2$

(a) $$\min_{\beta_1, \beta_2} 2\left[y - x(\beta_1 + \beta_2)\right]^2 + \lambda(\beta_1^2 + \beta_2^2)$$

(b) The RSS depends only on $\beta_1 + \beta_2$. For a fixed sum, the ridge penalty is minimized when $\beta_1 = \beta_2$.

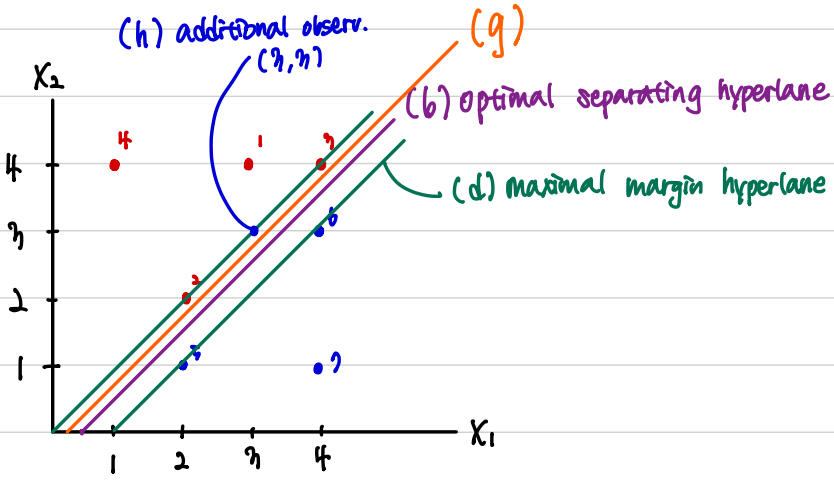
Therefore, for $\lambda > 0$, $\hat{\beta}_1 = \hat{\beta}_2$.

(c) $$\min_{\beta_1, \beta_2} 2\left[y - x(\beta_1 + \beta_2)\right]^2 + \lambda(|\beta_1| + |\beta_2|)$$

(d) Let t = the best value of $\beta_1 + \beta_2$. The predictions depend only on this sum. When $t > 0$, any two nonnegative coefficients that add up to t also have the same lasso penalty. The same applies when $t < 0$ and both coefficients are nonpositive. Therefore, if $t \neq 0$, there are many optimal pairs of coefficients.

ISLR 9.7.3

(a)



(b) $1 - 2X_1 + 2X_2 = 0$

$2X_2 = 2X_1 - 1$, $X_2 = X_1 - \frac{1}{2}$

(c) Classify to Red if $1 - 2X_1 + 2X_2 > 0$, and classify to Blue otherwise.

$\beta_0 = 1$, $\beta_1 = -2$, $\beta_2 = 2$

(d) $X_2 = X_1$, $X_2 = X_1 - 1$

(e) Observation 2, 3, 5, 6

(f) Observation 7 is not a support vector, so a slight movement of it will not affect the maximal margin hyperlane.

(g) $X_2 = X_1 - \frac{1}{4}$

(h) (2, 2)